

Your grade: 100%

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1. You have two corresponding points in two stereo images, and you want to find the epipolar line equation in Image 2 for a point in Image 1. The point in Image 1 is $(x_1, y_1) = (3, 4)$. You are given the fundamental matrix F as:

1 / 1 point

$$\begin{bmatrix} 0 & -1 & 2 \\ 1 & 0 & -3 \\ -2 & 3 & 0 \end{bmatrix}$$

Find the equation of the epipolar line in Image 2 corresponding to the point $(3, 4)$ in Image 1.

- ☒ $2x - 6 = 0$
- ☐ $2y - 2x - 6 = 0$
- ☐ $2x + 2y - 6 = 0$
- ☐ $2y - 6 = 0$

✔ Correct

To find the epipolar line, we need to use the Epipolar Constraint Equation:

$$\begin{bmatrix} u_l & v_l & 1 \end{bmatrix} \begin{bmatrix} f_{11} & f_{12} & f_{13} \\ f_{21} & f_{22} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{bmatrix} \begin{bmatrix} u_r \\ v_r \\ 1 \end{bmatrix} = 0$$

The left image point and the fundamental matrix are given.